

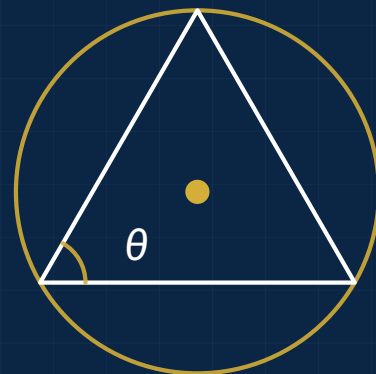


DR ESLAM AHMED
eliteigcse.com

CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 9

Area & Volume of Similar Shapes

Lengths $\times k$, areas $\times k^2$, volumes $\times k^3$. Always go back to the length factor k first, then up to what you need.



4

VARIANTS

18

PAST-PAPER PARTS

30

PRACTICE QUESTIONS

YOU WILL MASTER

- V1 The three scale factors
- V2 Area $\leftrightarrow$ volume
- V3 Sums & differences
- V4 Frustums & algebra

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Similar areas & volumes at a glance



Your notes

Area & Volume of Similar Shapes

THE FACTORS

V1 Three scale factors

 k, k^2, k^3

PRACTICE ONLY • BE READY

CONVERT

V2 Area $\leftrightarrow$ volume

 $\frac{300}{108} \rightarrow k \rightarrow k^3$

PAST PAPERS: 8 OF 18

HARDER

V3 Sums & differences

 $V_B - V_A = 1197$

PAST PAPERS: 4 OF 18

V4 Frustums & algebra

large – small cone

PAST PAPERS: 6 OF 18

THE KEY HABIT

BACK TO k FIRST

Area factor: $\sqrt{\quad}$. Volumefactor: $\sqrt[3]{\quad}$.

★ How often does each variant come up?

17 past-paper questions (Jan 2020 → Nov 2025) hold 18 parts; 30 practice parts — grouped by type.

	PAST PAPERS (OUT OF 18)	PRACTICE PARTS
V1 Three scale factors	0	18
V2 Area and volume	8	8
V3 Sums, differences	4	1
V4 Frustums, algebra	6	3

WHAT THIS TELLS YOU On almost every paper, near the end (Q17–26, 3–6 marks). Every question is the same chain: find k from the information you have, then square or cube it.

BEFORE YOU START

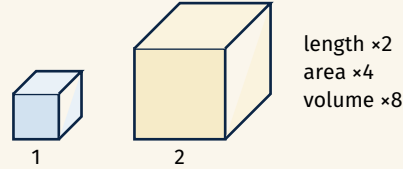
The big idea



Your notes

**Double the length — area $\times 4$,
volume $\times 8$.**

Area has **two** lengths multiplied,
volume has **three**. So if every length is
 $\times k$, area is $\times k^2$ and volume is $\times k^3$.



The three factors
Length k

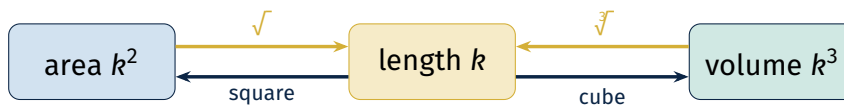
Heights, radii, edges.

Area k^2

Surface area, any area.

Volume k^3

Also mass, capacity.

Going between them
**CALCULATOR FIRST, THEN CHECK**

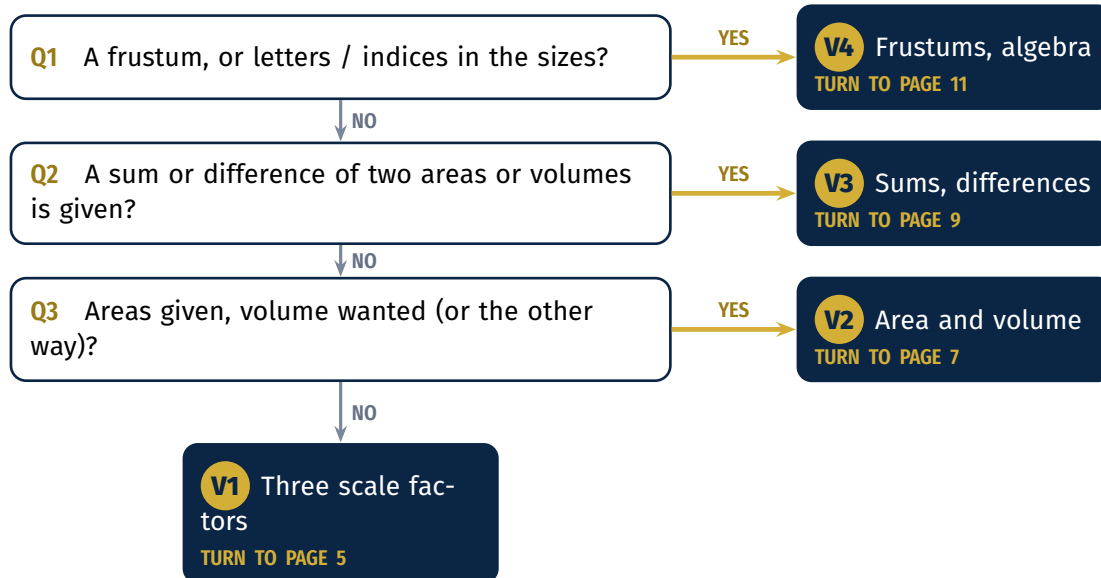
Keep k as a fraction ($\frac{5}{3}$) so k^2 and k^3 stay exact.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?


 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

The Three Scale Factors

Length k , area k^2 , volume k^3 ; find k from a pair of lengths.

SEEN IN PAST PAPERS

0 / 18

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 18
Your notes

? What is it?

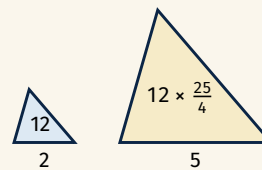
In plain words

Find k from **two matching lengths**.Areas use k^2 ; volumes use k^3 .Given a ratio of areas, **square-root** it for the lengths; a ratio of volumes, **cube-root** it.

IT LOOKS LIKE

lengths 2 : 5

areas 9 : 25



How do I do it?

1 k $\frac{\text{new length}}{\text{old length}}$ **2** Which quantity?

Length, area or volume?

3 Power k , k^2 or k^3 .**4** MultiplyOld value $\times$ the right factor. **k , k^2 , k^3** Practice only — but used in every question.

V1 The Three Scale Factors

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"lengths in the ratio 2 : 5" → $k = \frac{5}{2}$

"the curved surfaces are in the ratio 9 : 25" → **Lengths 3 : 5**

mass or capacity → **Like volume: k^3**

WORKED EXAMPLE DR ESLAM

Scale factors • 2 marks

Two similar shapes have lengths in the ratio 2 : 5. The small one has area 12 cm². Find the area of the large one.

STEP 1 $k = \frac{5}{2}$ ◀ length factor

STEP 2 $k^2 = \frac{25}{4}$ ◀ area factor

STEP 3 $12 \times \frac{25}{4} = 75$ ◀ multiply

ANSWER 75 cm²✓ 75 : 12 = 25 : 4 = 5² : 2² ✓**WATCH OUT**

$$\times 12 \times \frac{5}{2} = 30$$

$$\checkmark 12 \times \frac{25}{4} = 75$$

Area uses k^2 , not k .

EXAM TIP

Write "L k , A k^2 , V k^3 " at the top of the question.

Check yourself

calculator first • answers on page 13

1.1 Practice • Medium

Two similar cones have curved surface areas in the ratio 9 : 25. Ratio of their heights?

Answer:

1.2 Practice • Medium

Lengths are $\times 6$. The small shape has area 300 cm². Area of the large one?

Answer:

1.3 Dr Eslam

Two similar solids have volumes in the ratio 8 : 27. Ratio of their lengths?

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM**Area ↔ Volume**

Given areas: $\sqrt{\quad}$ to k , then cube. Given
volumes: $\sqrt[3]{\quad}$ to k , then square.

SEEN IN PAST PAPERS

8 / 18

PRACTICE QUESTIONS: 8


Your notes

Came up in: Oct/Nov 2020 P1H Q20 • Jan 2021 P1H Q20 • May 2021 P2H Q20 • Jan 2022 P1HR Q18 • Jun 2022 P1HR Q20 • Jun 2024 P2HR Q24 • Nov 2025 P2H Q23

? What is it?**In plain words**

You can't go straight from area to volume. Go

through the length factor: area factor $\rightarrow \sqrt{\quad} \rightarrow k \rightarrow$ cube $\rightarrow$ volume factor.

Or volume factor $\rightarrow \sqrt[3]{\quad} \rightarrow k \rightarrow$ square $\rightarrow$ area factor.

$$\frac{300}{108} = \frac{25}{9}$$

$$k = \frac{5}{3}$$

$$k^3 = \frac{125}{27}$$

$$135 \times \frac{125}{27} = 625$$

IT LOOKS LIKESA 108 $\rightarrow$ 300

V = 135

How do I do it?**1 Known factor**

Area or volume ratio.

2 Back to k $\sqrt{\quad}$ or $\sqrt[3]{\quad}$.**3 Up**

Square or cube for what you need.

4 MultiplyOld value $\times$ factor.**BACK TO k FIRST** The most common variant.

V2 Area ↔ Volume

VARIANT 2 • SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

"Shape R has surface area 108 and volume 135; S has surface area 300" → $\sqrt{\frac{300}{108}}$, then cube

"A has volume 1836; B has volume 4352 and surface area 1120" → $\sqrt[3]{\quad}$, then square

"20% less volume ... k% less surface area" → $0.8^{2/3}$

 WORKED EXAMPLE **PAST PAPER**

Oct/Nov 2020 • Paper 1H • Q20 • 3 marks

R and S are similar. R has surface area 108 cm² and volume 135 cm³; S has surface area 300 cm². Work out the volume of S.

STEP 1 $\frac{300}{108} = \frac{25}{9}$ ◀ area factor

STEP 2 $k = \sqrt{\frac{25}{9}} = \frac{5}{3}$ ◀ length factor

STEP 3 $k^3 = \frac{125}{27}$ ◀ volume factor

STEP 4 $135 \times \frac{125}{27} = 625$ ◀ multiply

ANSWER 625 cm³

✓ S is bigger, volume bigger ✓

 WATCH OUT

$$\times 135 \times \frac{25}{9}$$

$$\checkmark 135 \times \frac{125}{27}$$

The area factor is not the volume factor.

 EXAM TIPWrite the three lines: area factor, k , volume factor — 1 mark each. **Check yourself**

calculator first • answers on page 13

2.1 Jan 2021 P1H

A has volume 1836 cm³;
B has volume 4352 cm³
and surface area 1120
cm². Surface area of A?

Answer:

2.2 May 2021 P2H

Small block 300 cm³;
surface areas small :
large = 25 : 36. Volume
of a large block?

Answer:

2.3 Nov 2025 P2H

P: area 200 cm², volume
672 cm³. Q is similar
with area 450 cm².
Volume of Q?

Answer:

V3

VARIANT 3

Sums & Differences

Write the big one as (factor) $\times$ the small one; the sum or difference becomes one equation.

SEEN IN PAST PAPERS

4 / 18

PRACTICE QUESTIONS: 1


 Your notes

Came up in: Jan 2020 P2HR Q20 • Jan 2022 P1H Q17 • May 2024 P1HR Q18 • Jun 2025 P2H Q26

? What is it?

In plain words

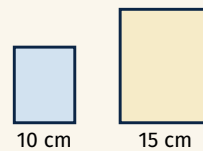
You are given $A + B$ or $B - A$, not the separate values. Write $B = k^3 A$ (or $k^2 A$), then the sum or difference has **one unknown**.

$$B - A = (k^3 - 1)A.$$

IT LOOKS LIKE

$$V_B - V_A = 1197$$

$$SA_A + SA_B = 1800$$



$$3.375V - V = 1197$$

How do I do it?

1 k

From the heights.

2 Factor k^2 (area) or k^3 (volume).**3** Equation $B = \text{factor} \times A$; substitute into the sum/difference.**4** Solve

Then answer the one asked.

 $B = k^3 A$, THEN SOLVE One unknown.

V3 Sums & Differences

VARIANT 3 • SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"The difference between the volume of A and the volume of B is 1197" → $(k^3 - 1)V_A = 1197$

"surface area of A + surface area of B = 1800" → $(1 + k^2)S_A = 1800$

"volume of A - volume of B = 735", A bigger → $(1 - k^3)V_A$ **with** $k < 1$

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 1H • Q17 • 4 marks

Similar vases: A has height 10 cm, B has height 15 cm. The volumes differ by 1197 cm³. Calculate the volume of A.

STEP 1 $k = \frac{15}{10} = 1.5$, $k^3 = 3.375$ ◀ volume factor

STEP 2 $V_B = 3.375V_A$ ◀ B in terms of A

STEP 3 $3.375V_A - V_A = 2.375V_A = 1197$ ◀ difference

STEP 4 $V_A = 504$ ◀ solve

ANSWER 504 cm³✓ $1701 - 504 = 1197$ ✓**WATCH OUT**✗ $1197 \div 3.375$ ✓ $1197 \div 2.375$ The difference is $(k^3 - 1)$ lots of V_A .**EXAM TIP**

Check by finding both volumes and subtracting.

Check yourself

calculator first • answers on page 13

3.1

Jan 2020 P2HR

Similar vases of heights 9 cm and 13 cm; total surface area 1800 cm². Surface area of the small vase A?

Answer:

3.2

May 2024 P1HR

Heights 30 cm (A) and 12 cm (B); surface area A - B = 178.5 cm². Surface area of A?

Answer:

3.3

Jun 2025 P2H

Heights 31 cm (A) and 18.6 cm (B); volume A - B = 735 cm³. Volume of A?

Answer:

V4

VARIANT 4

Frustums & Algebra

Frustum = large – small similar solid; with letters, keep everything in terms of k or x .

SEEN IN PAST PAPERS

6 / 18

PRACTICE QUESTIONS: 3



Your notes

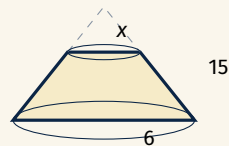
Came up in: Jan 2020 P1HR Q20 • Jun 2022 P1HR Q20 • Jun 2022 P2HR Q24 • May 2023 P1H Q23 • Nov 2023 P1H Q23 • Nov 2023 P2H Q23 • Jun 2025 P1HR Q26

? What is it?**In plain words**

A **frustum** is a cone (or pyramid) with the top cut off. The cut-off top is **similar** to the whole: its sizes are a fraction of the big one.

Frustum volume = large – small.

Algebra questions: same steps with letters.

**IT LOOKS LIKE**

$$\text{frustum } \frac{4212}{25} \pi$$

$$r = \frac{7h}{16}$$

How do I do it?**1 Two solids**

Large and small (removed) – similar.

2 Small sizes

Scale the large one by k .

3 Subtract

Frustum = large – small.

4 Solve

Equation in x (or k).

LARGE – SMALL The small top is similar.

V4 Frustums & Algebra

VARIANT 4 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"The frustum is made by removing a small cone from a similar large cone" → **Large – small**

"Given the volume of the frustum is $\frac{4212}{25}\pi$ " → **Equation in x**

WORKED EXAMPLE PAST PAPER

November 2023 • Paper 2H • Q23 • 5 marks

Frustum: a large cone (height 15 cm, radius 6 cm) minus a similar small cone of radius x cm. Its volume is $\frac{4212}{25}\pi \text{ cm}^3$. Find x .

STEP 1 $\frac{1}{3}\pi(6)^2(15) = 180\pi$ ◀ large cone

STEP 2 $h = \frac{15}{6}x = 2.5x$; $\frac{1}{3}\pi x^2(2.5x) = \frac{5}{6}\pi x^3$ ◀ small cone

STEP 3 $180 - \frac{5}{6}x^3 = \frac{4212}{25} \Rightarrow x^3 = 13.824$ ◀ frustum = large – small

STEP 4 $x = 2.4$ ◀ cube root

ANSWER $x = 2.4$

✓ $180\pi - 11.52\pi = 168.48\pi$ ✓

WATCH OUT

✗ small cone height $15 - x$

✓ $\frac{15}{6}x$

The small cone is similar: height and radius scale together.

EXAM TIP

Divide through by π early – the numbers get simpler.

Check yourself

calculator first • answers on page 13

4.1 Jan 2020 P1HR

Frustum: cone (height $2h$, radius r) minus a similar cone of height h ; volume = sphere of radius r . Find r .

Answer:

4.2 Jun 2022 P1HR

Statue B has 20% less volume than similar statue A. B's surface area is $k\%$ less. Find k .

Answer:

4.3 Nov 2023 P1H

A: height 3^x , area 7776, volume 8^x . B: area 486, volume 2^{x+4} . Height of B?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 $3 : 5$

1.2 $10\,800\text{ cm}^2$

1.3 $2 : 3$

2.1 630 cm^2

2.2 518.4 cm^3

2.3 2268 cm^3

3.1 583.2 cm^2

3.2 212.5 cm^2

3.3 937.5 cm^3

4.1 $r = \frac{7h}{16}$

4.2 $k = 13.8$

4.3 60.75 cm

THE WHOLE TOPIC ON ONE PAGE

Similar shapes cheat sheet



Your notes
V1 Factors k, k^2, k^3

Length, area, volume.

V2 Area ↔ volume 8/18

$$\frac{25}{9} \rightarrow \frac{5}{3} \rightarrow \frac{125}{27}$$

Back to k first.**V3 Sum, difference 4/18**

$$(k^3 - 1)V = 1197$$

One unknown.

V4 Frustum 6/18

large – small

The top is similar.

**TOP 3 LOST MARKS**

1. area factor used for volume ×
2. dividing a difference by k^3 ×
3. frustum top not scaled ×

BOTTOM LINE L $k \cdot A k^2 \cdot V k^3$ · back to k before going up.